(2)

(Please write your Enrollment Number) Enrollment No. 11101172022

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OffLine)

Course Name: B.Tech, Semester: 6

(May, 2025)

Subject Code: BAI 322	Subject: Data Warehousing and Business Intelligence
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) Differentiate between a database system and a data warehouse.		CO1
	b) What is ETL? Explain each component.		CO1
	c) Define star schema with an example.		CO2
	d) What is OLAP? List its types and key characteristics.		CO2
	e) Define data mining and explain the role of KDD.		CO3
	f) Discuss the importance of data mining in business decision-making.		CO3
	g) Write a short note on outlier analysis.		CO4
	h) What are the benefits of using association rule mining?		CO4
	UNIT I		CO Mapping
Q2	Explain the architecture of a data warehouse. Discuss the three-tier architecture and the role of metadata in detail.	(10)	CO1
Q3	Discuss data pre-processing in detail. Explain data cleaning and data transformation with suitable examples.	(10)	CO1
	UNIT II		CO Mapping
Q4	Define dimensional modeling. Explain STAR, Snowflake, and Fact Constellation schemas with appropriate diagrams.	(10)	CO2
Q5	What are OLAP operations? Describe with suitable examples: drill-down, roll-up, slice, dice, and pivot.	(10)	CO2
	UNIT III		CO Mapping
Q6	Differentiate classification and prediction in data mining. How does decision tree induction work? Illustrate with an example.	(10)	CO3
Q7	What is KNN algorithm? Explain its working and use cases. Compare it with Decision Tree classification.	(10)	CO3
			CO Mapping
Q8	What is clustering in data mining? Explain the K-Means algorithm with an example. Mention its limitations.	(10)	CO4
Q9	What will be the association rules using Apriori Algorithm: Given the following transactions: TID Items T1 A, B, C T2 A, C T3 A, D T4 B, E T5 A, B, C, E T6 B, C Support : 50% Confidence: 75%	(10)	CO4